

year

2021

2023

Avg PM10 ($\mu\text{g}/\text{m}^3$)

34.3

Avg PM2.5 ($\mu\text{g}/\text{m}^3$)

17.3

PM10 hours over KR standard (%)

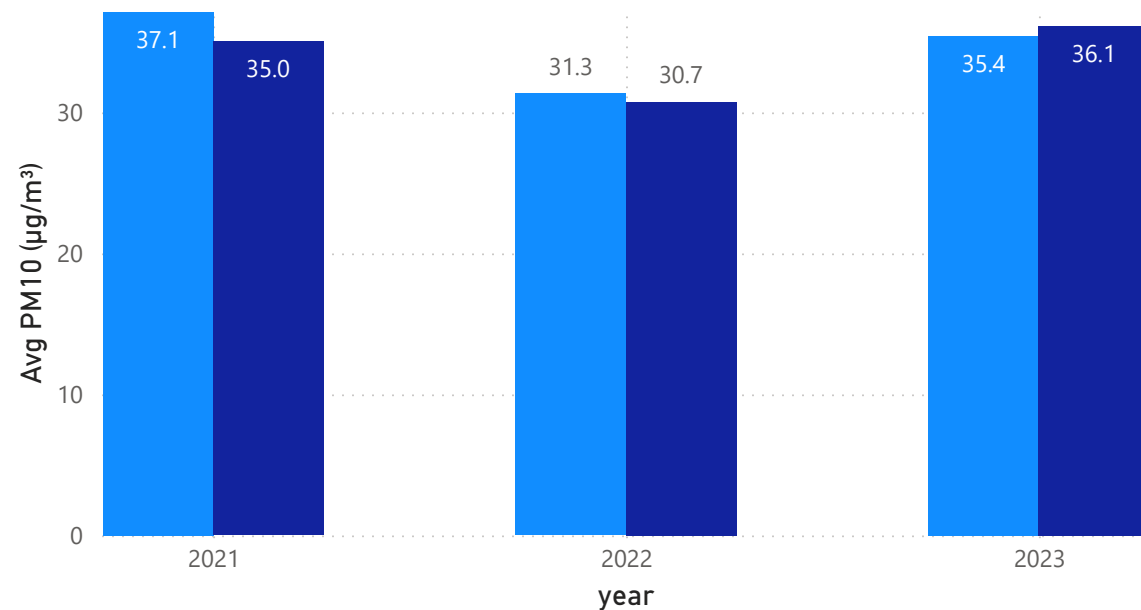
16.5%

PM2.5 hours over KR standard (%)

44.1%

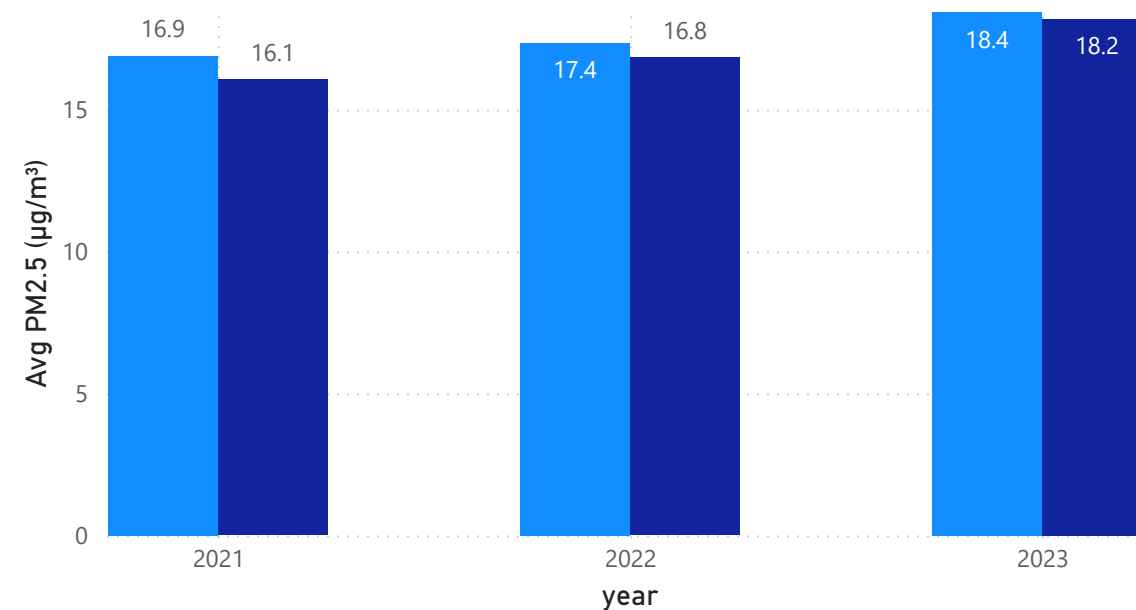
Avg PM10 by year & area ($\mu\text{g}/\text{m}^3$)

Area (East / West) ● 동부권역 ● 서부권역



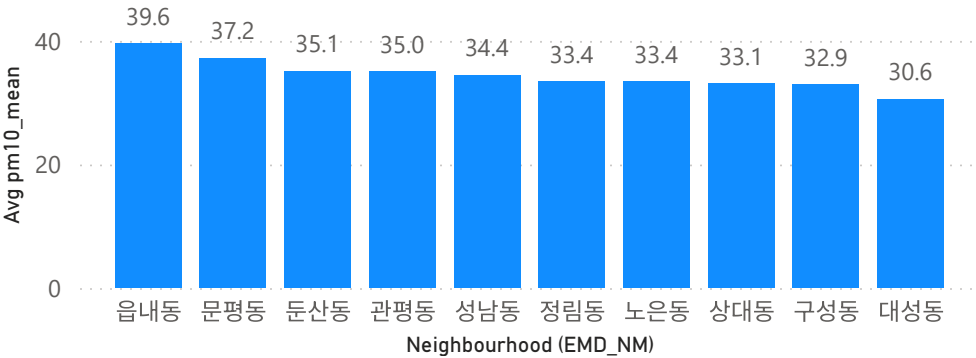
Avg PM2.5 by year & area ($\mu\text{g}/\text{m}^3$)

Area (East / West) ● 동부권역 ● 서부권역

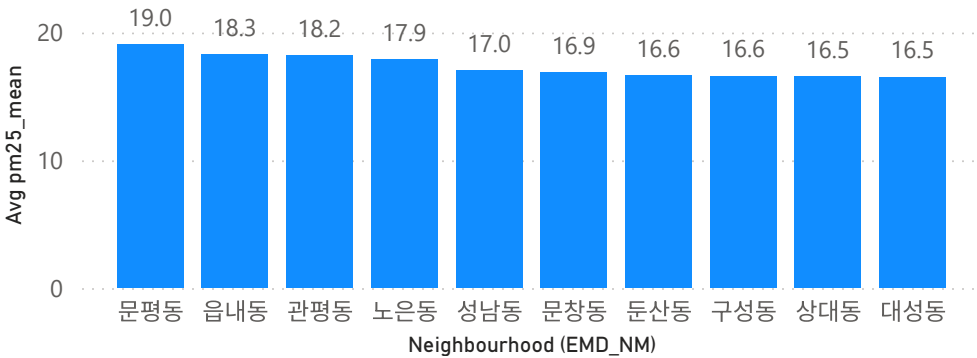


- season
- Fall
- Spring
- Summer
- Winter

Avg PM10 by neighbourhood (µg/m³)



Avg PM2.5 by neighbourhood (µg/m³)



Neighbourhood (EMD_NM)	# of hours	year	season	Avg PM10 (µg/m³)	Avg PM2.5 (µg/m³)	PM10 90th percentile (µg/m³)	PM2.5 90th percentile (µg/m³)	Share of hours over PM10 standard (%)	Share of hours over PM2.5 standard (%)
대성동	2149	2022	Summer	15.3	8.4	26.0	16.0	0.00	0.12
상대동	2132	2022	Summer	18.7	8.4	31.0	17.0	0.01	0.12
관평동	2150	2022	Summer	19.1	8.1	31.0	17.0	0.01	0.13
둔산동	2114	2022	Summer	19.0	9.0	34.0	19.0	0.01	0.15
성남동	2151	2022	Summer	19.6	10.1	33.0	21.0	0.01	0.20
읍내동	2135	2022	Summer	21.1	10.7	35.0	20.0	0.01	0.21
구성동	2150	2022	Summer	18.6	10.6	32.0	22.0	0.00	0.22
상대동	2014	2021	Summer	19.5	10.3	40.0	23.0	0.04	0.22
노은동	2143	2022	Summer	18.0	11.4	31.0	21.0	0.00	0.24
합계				34.1	17.2	59.7	33.1	0.16	0.44

PM10 alert hours (total)

1522

Total hours (2021–2023)

Neighbourhood (EMD_NM)	Avg PM10 during PM10 alerts (µg/m³)	Avg PM10 during PM2.5 alerts (µg/m³)	PM10 alert hours (#)
대성동	161.04	42.72	152
상대동	166.79	43.29	127
문창동	173.19	36.91	152
정림동	177.07	34.83	127
구성동	178.43	39.90	127
문평동	181.34	36.65	152
읍내동	181.82	37.55	152
성남동	182.11	37.16	152

PM2.5 alert hours (total)

2054

Total hours (2021–2023)

Neighbourhood (EMD_NM)	Avg PM10 during PM2.5 alerts (µg/m³)	Avg PM2.5 during PM2.5 alerts (µg/m³)	PM2.5 alert hours (#)
관평동	169.21	87.41	184
구성동	168.27	65.66	184
노은동	166.25	74.11	184
둔산동	165.68	77.02	184
상대동	161.12	77.89	184
정림동	169.76	72.38	184
대성동	193.57	82.69	190
문창동	215.60	75.18	190

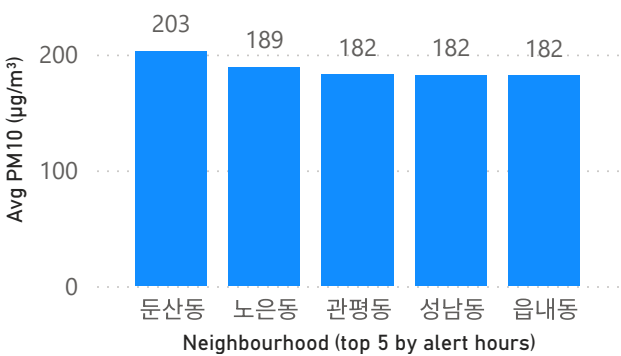
Dust advisory hours (total)

308

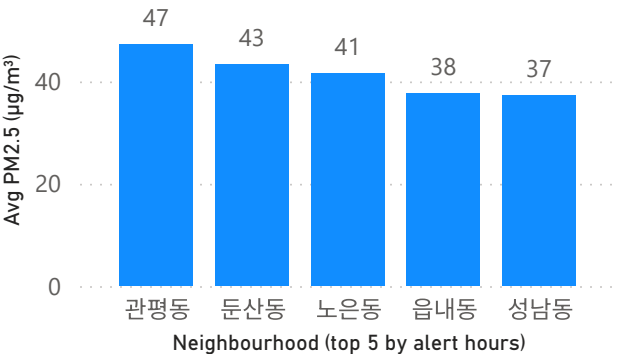
Total hours (2021–2023)

Neighbourhood (EMD_NM)	Avg PM10 during dust advisories (µg/m³)	Avg PM2.5 during dust advisories (µg/m³)	Dust advisory hours (#)
관평동	19.1	8.1	2150
관평동	19.8	11.8	2089
관평동	19.8	12.3	2147
관평동	23.2	14.2	2174
관평동	29.2	16.7	2145
관평동	31.4	13.6	2159
관평동	38.7	17.1	2199
관평동	41.8	24.3	2140

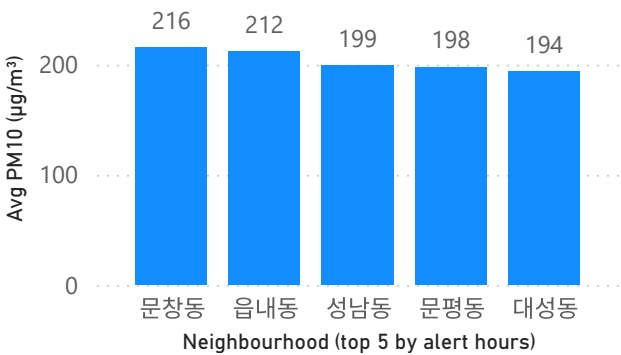
Avg PM10 during PM10 alerts (µg/m³)



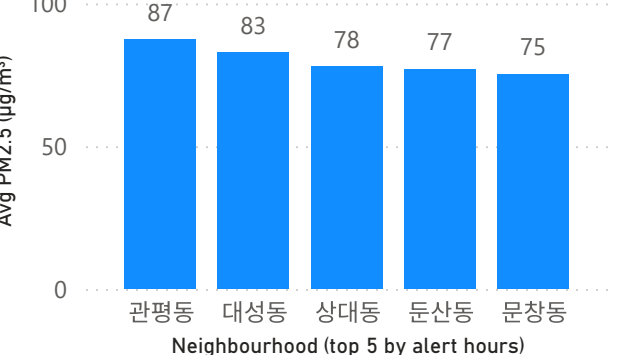
Avg PM2.5 during PM10 alerts (µg/m³)



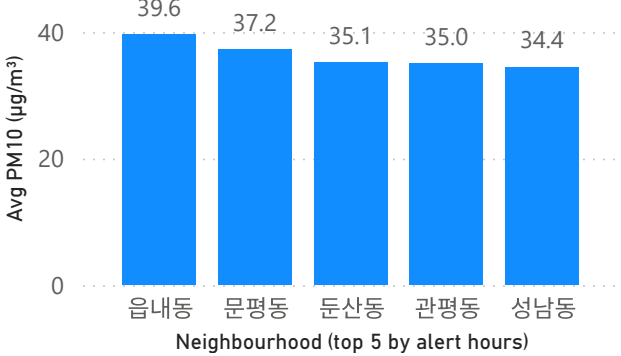
Avg PM10 during PM2.5 alerts (µg/m³)



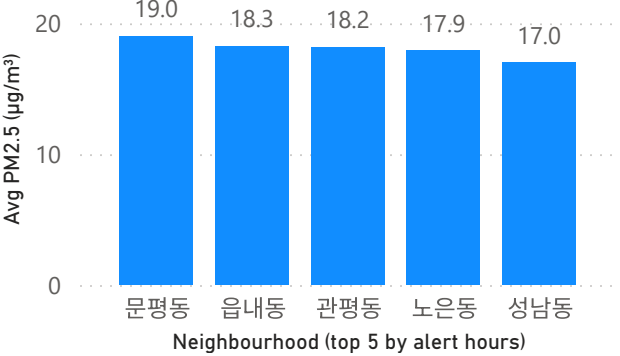
Avg PM2.5 during PM2.5 alerts (µg/m³)



Avg PM10 during dust advisories (µg/m³)



Avg PM2.5 during dust advisories (µg/m³)



Target pollutant

pm10

pm25

Model type

rf

xgb

Data subset

all

dust

loc_alert

no_dust

no_loc_alert

dust: periods with national dust advisories

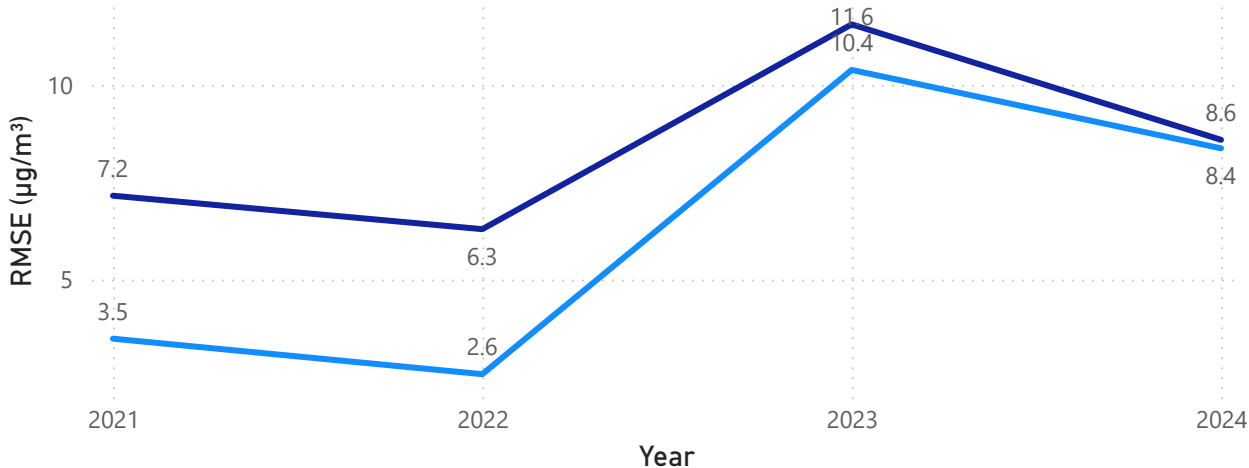
loc_alert: hours under local PM alerts

no_dust / no_loc_alert: non-event periods

Year	Subset	Target pollutant	Model	# of samples	RMSE (µg/m³)	MAE (µg/m³)	R²
2021	all	pm10	rf	92658	3.48	1.93	0.99
2022	all	pm10	rf	94423	2.57	1.71	0.99
2021	all	pm10	xgb	92658	7.15	4.79	0.98
2022	all	pm10	xgb	94423	6.29	4.39	0.91
2023	all	pm10	rf	94305	10.38	5.55	0.90
2023	all	pm10	xgb	94305	11.55	5.70	0.87
2024	all	pm10	rf	93690	8.37	4.70	0.86
2024	all	pm10	xgb	93690	8.59	4.68	0.85
합계				750152	7.30	4.18	0.92

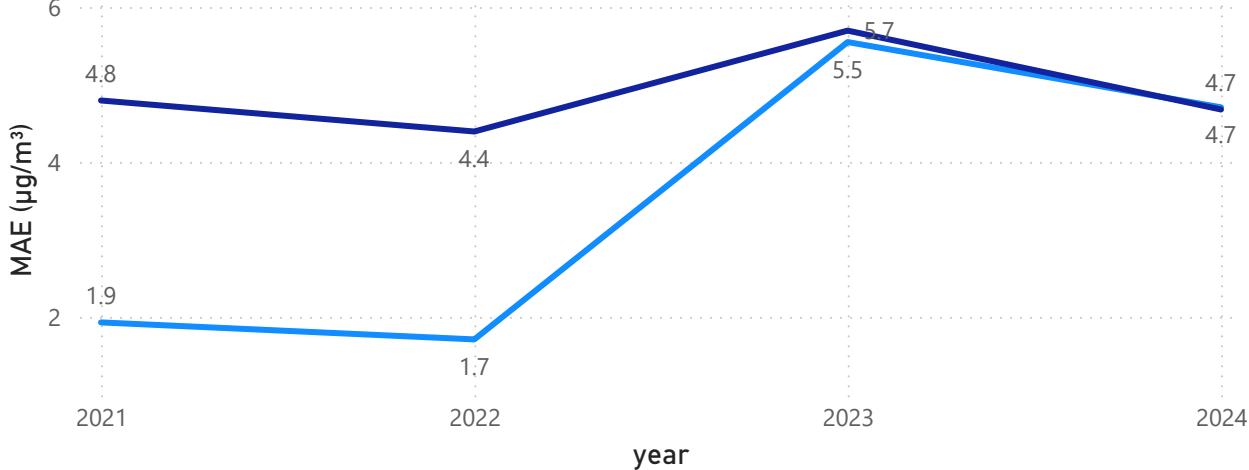
Avg RMSE by year & model (µg/m³)

Model (RF / XGB) rf xgb

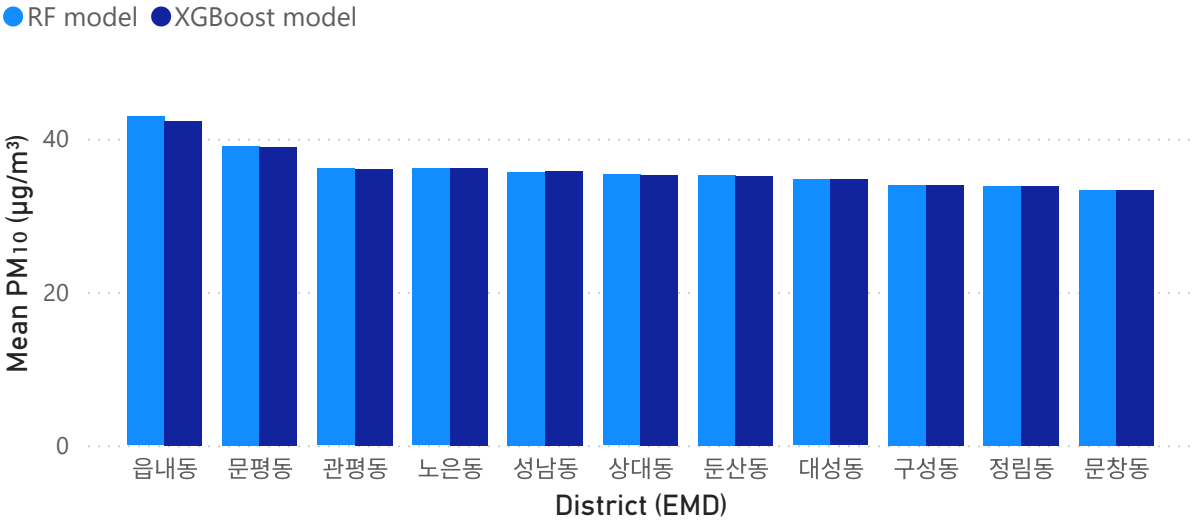


Avg MAE by year & model (µg/m³)

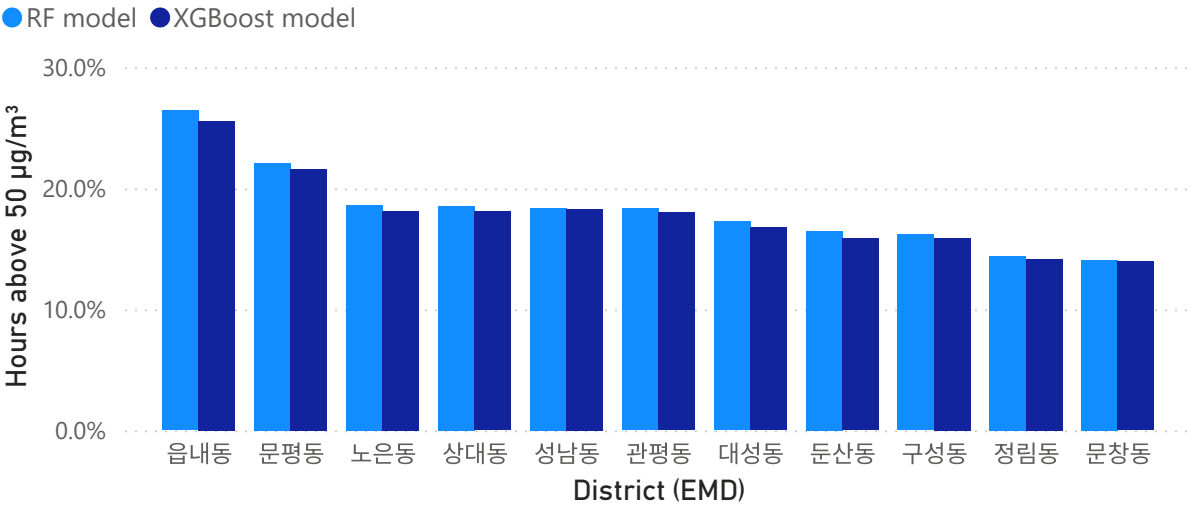
Model (RF / XGB) rf xgb



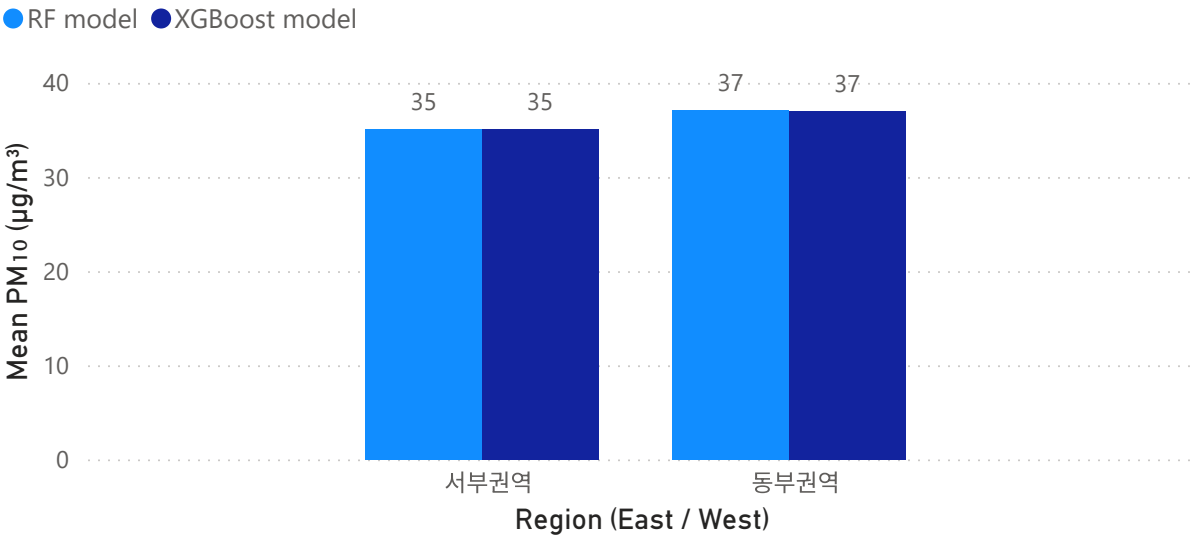
Mean PM₁₀ concentration in 2026 by district (RF vs XGBoost)



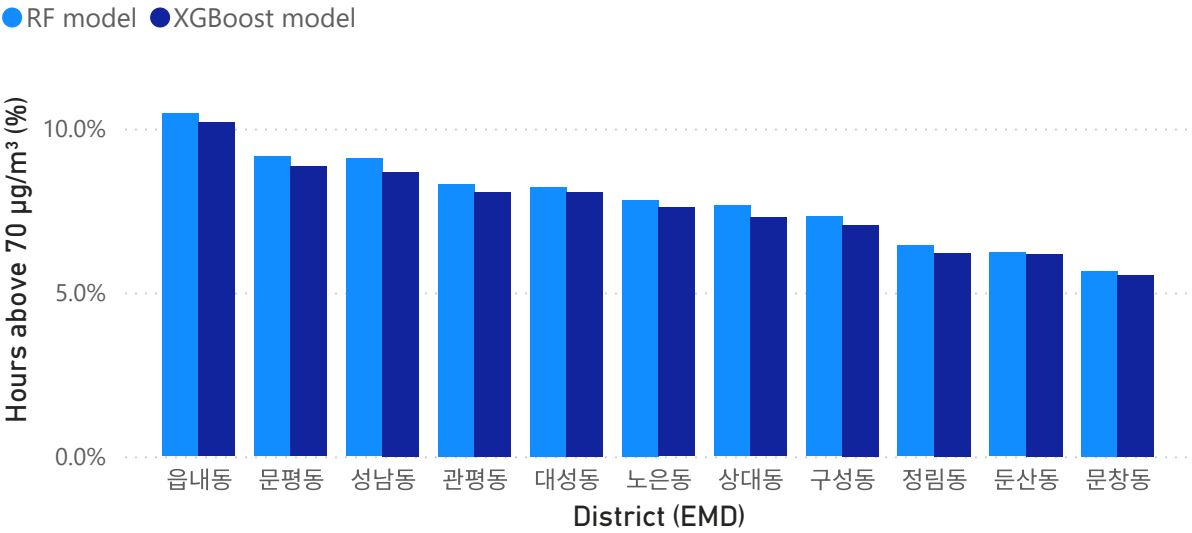
Fraction of hours with PM₁₀ ≥ 50 µg/m³ in 2026 (RF vs XGBoost)



Mean PM₁₀ in 2026 by East/West region (RF vs XGBoost)

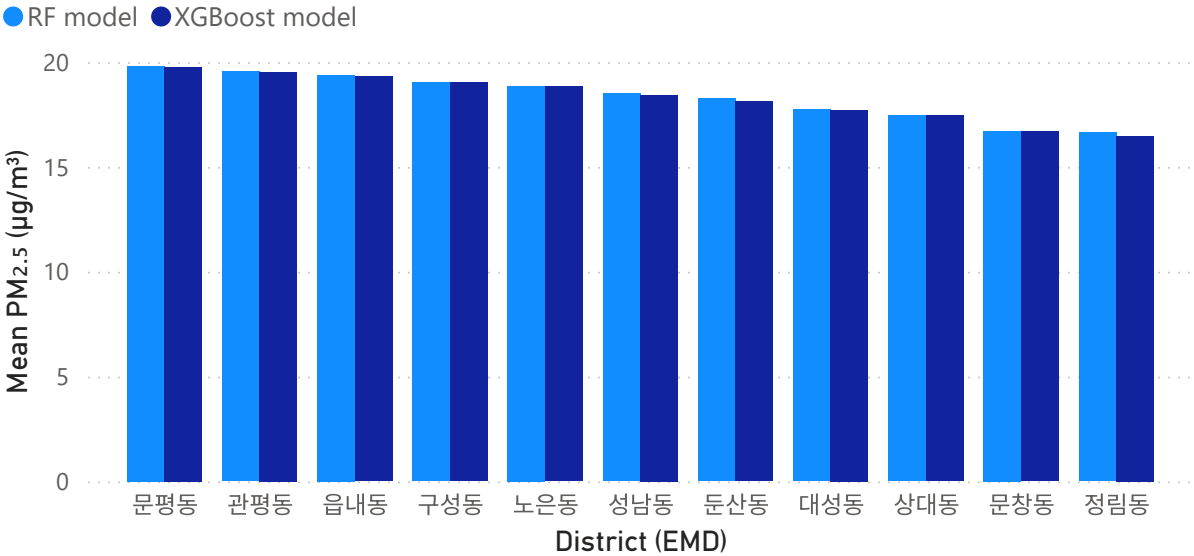


Fraction of hours with PM₁₀ ≥ 70 µg/m³ in 2026 (RF vs XGBoost)

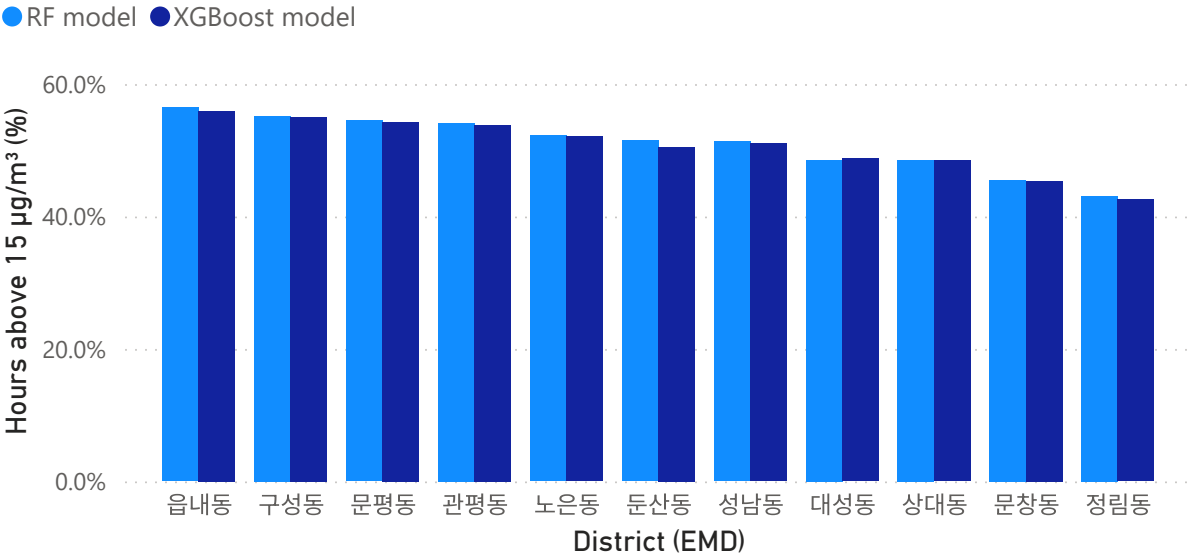


Scenario 2026 – PM₁₀: district-level and East/West mean, exceedance (≥50 µg/m³) and high-episode (≥70 µg/m³) fractions by RF vs XGBoost for analogue scenarios A–C.

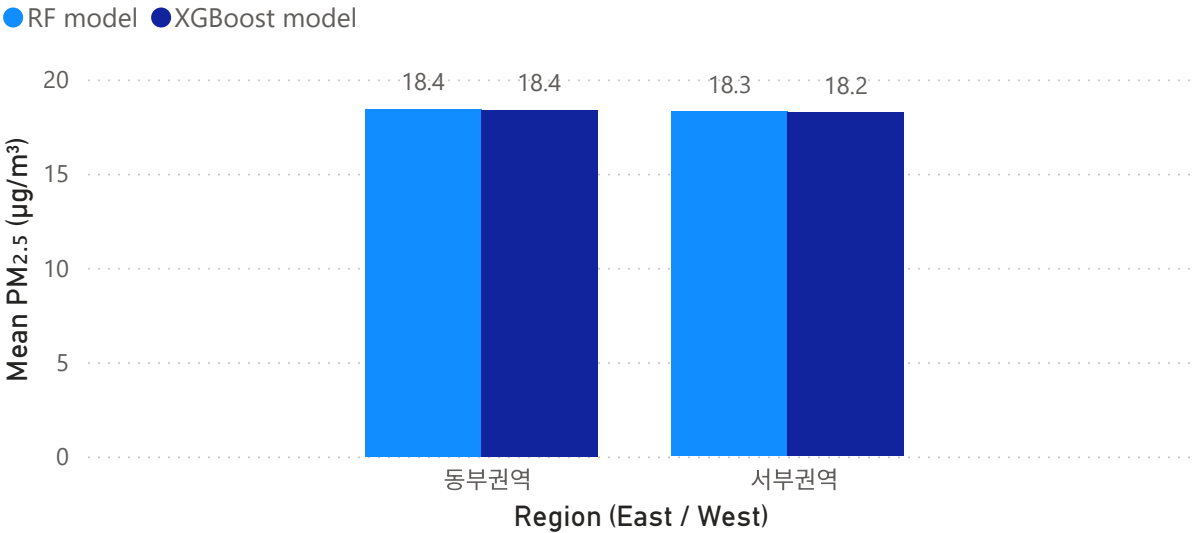
Mean PM_{2.5} concentration in 2026 by district (RF vs XGBoost)



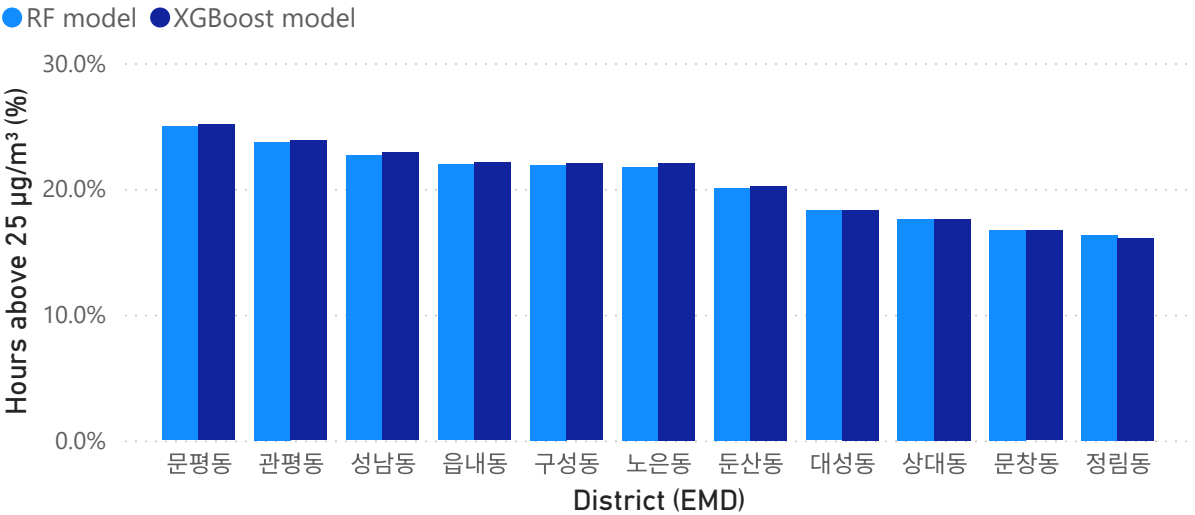
Fraction of hours with PM_{2.5} $\geq 15 \mu\text{g}/\text{m}^3$ in 2026 (RF vs XGBoost)



Mean PM_{2.5} in 2026 by East/West region (RF vs XGBoost)



Fraction of hours with PM_{2.5} $\geq 25 \mu\text{g}/\text{m}^3$ in 2026 (RF vs XGBoost)



Scenario 2026 – PM_{2.5}: district-level and East/West mean, exceedance ($\geq 15 \mu\text{g}/\text{m}^3$) and high-episode ($\geq 25 \mu\text{g}/\text{m}^3$) fractions by RF vs XGBoost for analogue scenarios A–C.

0

Stations classified as high-risk for PM10 (ensemble)

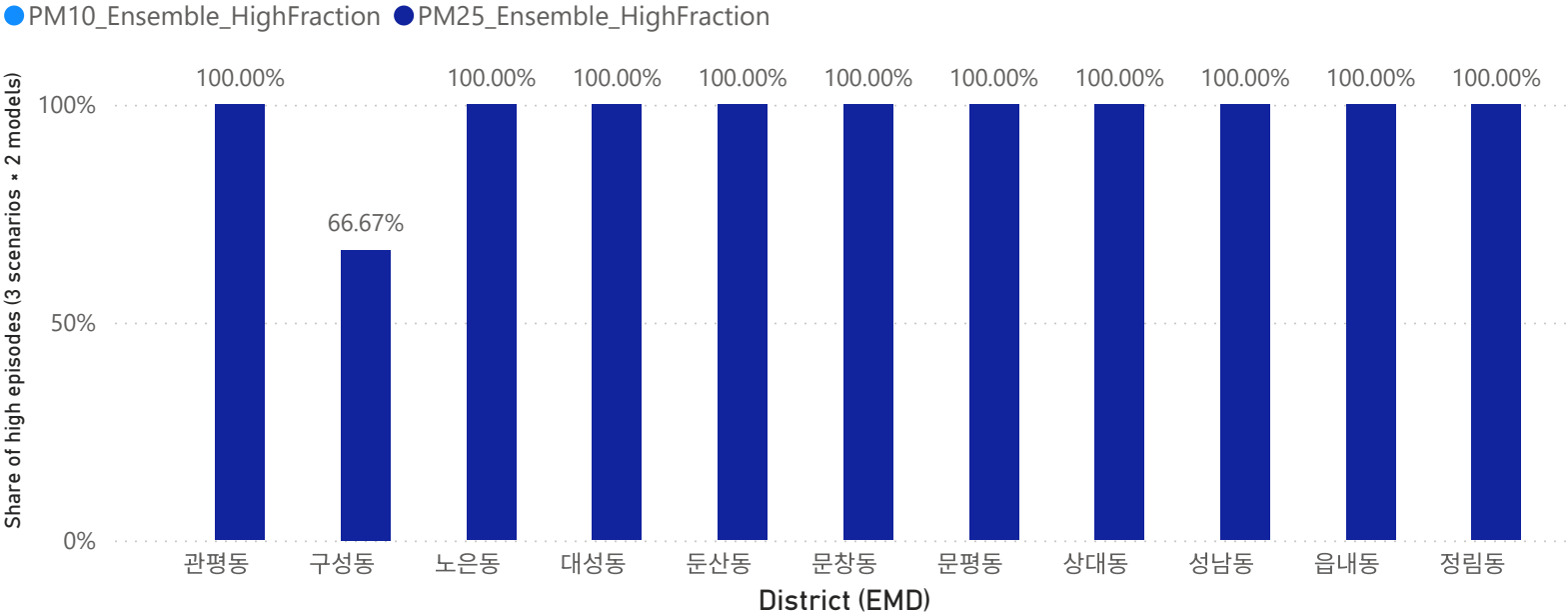
11

Stations classified as high-risk for PM2.5 (ensemble)

0

Stations high-risk for both PM10 and PM2.5

Share of high episodes in 2026 ensemble (PM₁₀ vs PM_{2.5}, by district)



District (EMD)	Total high-episode hits (PM10, all models)	Share of high episodes (PM10, ensemble)	Total high-episode hits (PM2.5, all models)	Share of high episodes (PM2.5, ensemble)
관평동	0	0.00%	6	100.00%
노은동	0	0.00%	6	100.00%
대성동	0	0.00%	6	100.00%
둔산동	0	0.00%	6	100.00%
문창동	0	0.00%	6	100.00%
문평동	0	0.00%	6	100.00%
상대동	0	0.00%	6	100.00%
성남동	0	0.00%	6	100.00%
읍내동	0	0.00%	6	100.00%
정림동	0	0.00%	6	100.00%
구성동	0	0.00%	4	66.67%
합계	0	0.00%	64	96.97%

Scenario 2026 – ensemble (Part 2): station-level ensemble risk scores for PM₁₀ and PM_{2.5} across all analogue scenarios (A–C) and both models, highlighting monitoring sites that remain high-risk under multiple futures.